

ES Futures Volatility: Distribution, Intraday Seasonality and Persistence

An empirical study of E-mini S&P 500 futures from one-second best-bid/offer data, November 2021 – September 2026

Written up from the notebook *General ES Vol AutoCorr.ipynb* · 13 September 2026

Summary. This report characterises the volatility of the front-month E-mini S&P 500 future (ES) using roughly 89 million one-second snapshots of the top of the order book. It answers four questions: how ES returns are distributed at horizons from 30 seconds to one day; how volatility varies through the 23-hour trading day and the week; how quickly the extra volatility of a market open or a shock fades; and how persistent (and therefore forecastable) volatility is. The headline findings are: (i) annualised realised volatility is about 16.7% in trading time and is flat across horizons, while excess kurtosis falls from about 190 at 30–60 seconds to 13 at one day – heavy tails at every horizon; (ii) the intraday profile is enormously uneven: the 09:30–10:30 ET cash-open hour carries 15% of all daily variance in 4% of the minutes, and the single minute after the 18:00 ET Globex reopen and the 08:30 ET data release run at annualised vols of 149% and 110%; (iii) the excess vol of an open fades with a half-life of only 3–7 minutes, but a genuine vol shock does not – conditional variance is still 8–9 times its baseline six and a half hours after a top-1% unscheduled minute; (iv) once the deterministic time-of-day pattern is removed, volatility has long memory: the autocorrelation of one-minute realised variance is well described by a fast component with a half-life of about 80 minutes plus a slow component with a half-life of roughly eight trading days, and a HAR regression on hourly data explains about 70–74% of the variance of next-hour to next-day log realised variance out of sample. The final section discusses what this implies for a short-horizon volatility model.

1. Data and definitions

The raw material is Databento's *bbo-1s* feed for the continuous front-month contract ES.c.0 from 1 November 2021 to 11 September 2026, stored as five yearly Parquet files (about 18.8 million rows in the first year alone, and roughly 89 million rows in total). Each row is a once-per-second snapshot of the best bid price and size, the best ask price and size, the number of orders at each, the last trade, and the instrument identifier – which changes when the front month rolls to the next quarterly contract. The feed omits seconds in which the book did not change, so a two- or three-second gap between rows is normal and simply means nothing happened.

Two price series are built from each snapshot and used throughout:

Mid

The plain midpoint, $(\text{bid} + \text{ask})/2$. ES trades in ticks of 0.25 index points, so the mid can only sit on a grid of half-ticks (0.125 points). At an index level near 5,000 one half-tick is 2.5 basis points, which is about the size of a typical 30-second move; consequently a large fraction of short-horizon mid returns are exactly zero, and the rest are multiples of a fixed step.

Size-weighted mid (swmid, also called the microprice)

$(\text{bid size} \times \text{ask price} + \text{ask size} \times \text{bid price}) / (\text{bid size} + \text{ask size})$. The mid is tilted towards the side with the *smaller* queue: when few contracts remain on the offer and many on the

bid, the next price change is more likely to be upward, and the microprice sits closer to the ask. It moves continuously as queue sizes change, so it is almost never exactly unchanged between snapshots, and it is a better estimate of the "true" price than the mid at very short horizons.

Log return

The difference in the natural logarithm of the price between two grid points. Log returns add across time, which is what makes the variance-scaling arguments below clean.

Realised volatility and realised variance

The realised variance over a window is the sum of squared log returns inside it; realised volatility is its square root. It is a direct measurement of how much the price actually moved, as opposed to an implied or model-based volatility. When it is quoted "annualised", the per-period figure has been scaled up to a one-year equivalent (see the box below).

Skewness and excess kurtosis

Skewness measures asymmetry (positive means the right tail is longer or heavier). Kurtosis measures how heavy both tails are relative to the centre; *excess* kurtosis subtracts the value 3 that a normal distribution has, so 0 means normal tails and large positive values mean that rare, very large moves carry a disproportionate share of the variance. Both statistics are dominated by the most extreme observations, so at short horizons with hundreds of millions of tiny moves and a few very large ones they should be read as orders of magnitude, not precise numbers.

Annualisation in trading time. Every "annualised" number in this report scales a per-period statistic by the number of such periods that actually occur in a year of *trading*, not by the number of calendar periods. In the cross-horizon table the scale factor is measured empirically as (number of returns in the sample) $\div$ (years spanned); in the intraday analysis it is 252 trading days $\times$ 1,380 trading minutes per 23-hour Globex day. Variance is multiplied by this factor and volatility by its square root. The convention treats a minute at 3 a.m. and a minute at 10 a.m. as the same amount of "time", and it treats closed periods (the daily 17:00–18:00 ET maintenance break, weekends, holidays) as non-existent. This is the right convention for a model that will be applied minute by minute while the market is open, and it is the reason the flat parts of the intraday curve line up with the 16.7% figure from the horizon table.

The one detail of the trading calendar that matters most for what follows is CME's daily maintenance break: Globex trading in ES halts from 16:00 to 17:00 Central time, which is 17:00–18:00 US Eastern time and 21:00–22:00 UTC (22:00–23:00 UTC in winter). The session that reopens at 18:00 ET is the start of the next trading day, running through the Asian and European hours into the US cash session (09:30–16:00 ET) and closing at 17:00 ET. Weekend trading stops on Friday at 17:00 ET and resumes on Sunday at 18:00 ET.

2. A first look at the one-second data

The study opens with the first million rows of the first file – roughly the three weeks from 1 to 22 November 2021 – to get a feel for the data before scaling up. Figure 1 plots the last-trade price over an eleven-hour stretch on the first day. The most obvious feature is the flat segment with no updates between roughly 21:00 and 22:00 UTC: this is the maintenance break described above, not missing data. The other feature is the texture: the price path is visibly "jittery" at the one-tick level even while it trends, which is the bid-ask bounce that a microprice is designed to smooth.

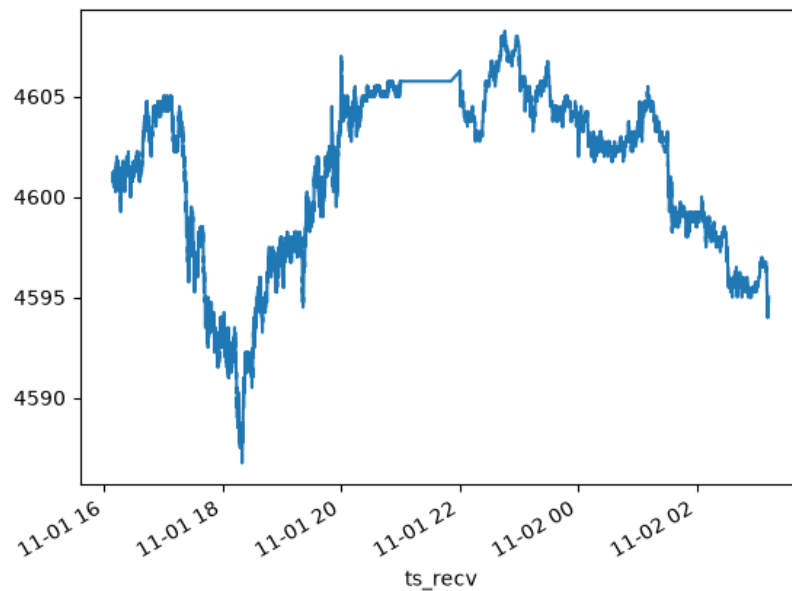


Figure 1. Last-trade price of ES.c.0 for rows 50,000–80,000 of the first row group (1 November 2021, about 16:00 UTC to 03:00 UTC the next day). The flat segment around 21:00–22:00 UTC is the CME maintenance break (17:00–18:00 ET) during which no quotes are published.

One-second log returns of the size-weighted mid were then computed over this block and their distribution examined (Figure 2). Note that the histogram is on a linear scale and is clipped to ± 2.5 basis points, so the tails are not visible at all; nearly everything sits within ± 1 basis point. The summary statistics for this three-week block are:

Mean one-second log return	1.9×10^{-8} (indistinguishable from zero)
Annualised volatility	14.3%, scaling the one-second standard deviation by $\sqrt{(365 \times 86,400)}$ calendar seconds
Excess kurtosis	5,191
Skewness	−15.7

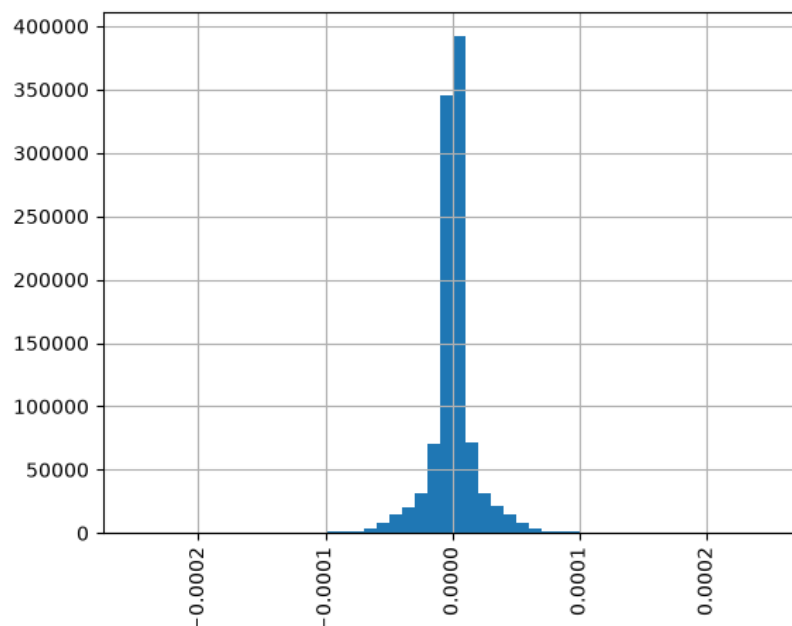


Figure 2. Histogram of one-second log returns of the size-weighted mid for the first row group (about 1.05 million observations), clipped to ± 0.00025 (± 2.5 bp). Linear frequency axis.

Two things about these first numbers are worth flagging because they are corrected later in the study. First, the 14.3% vol uses a *calendar*-second scale factor (31.5 million seconds per year) where the trading-time convention adopted afterwards would use about 20.9 million, and the "one-second" returns are between consecutive rows, which are not all one second apart because unchanged seconds are omitted from the feed; the figure is therefore only indicative, and it refers to a single three-week window in November 2021 rather than the full sample. Second, the excess kurtosis of over 5,000 and the skew of -16 are what one-second returns look like when the sample contains a few moves that are hundreds of times the typical one-second move – at this resolution a single large print dominates the fourth moment. These numbers are not wrong, but they are not stable either, and the cross-horizon analysis in Section 4 handles them more carefully.

3. One-minute sampled returns over the full sample

The second step loads the entire 2021–2026 dataset sampled on a 60-second grid (one row per minute, using the book as it stood at that instant), computes the size-weighted mid and its one-minute log returns, and looks at the distribution again (Figure 3). Over about 1.3 million one-minute returns the skewness is $+0.56$ and the excess kurtosis 314. Figure 3 is clipped at ± 20 basis points; the small bars at the edges are the returns beyond the clip, which have been piled into the outermost bins.

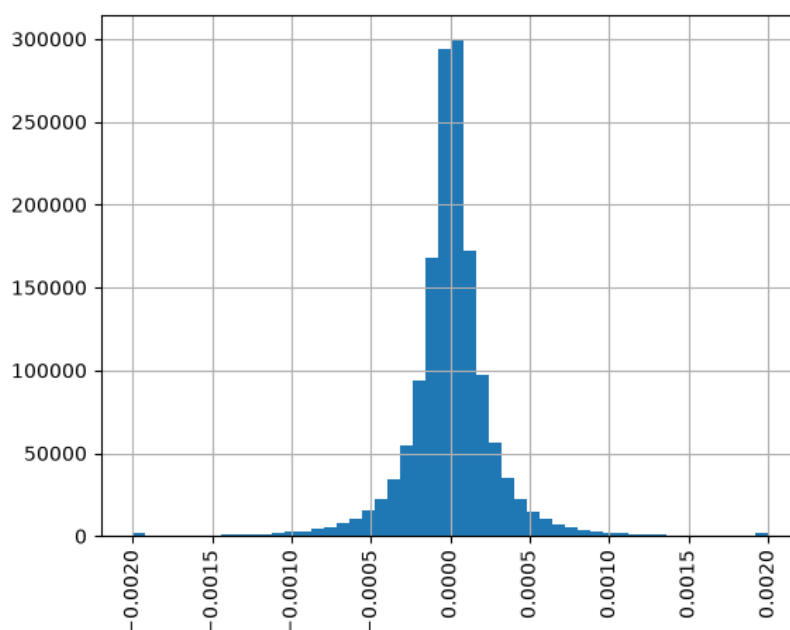


Figure 3. Histogram of one-minute log returns of the size-weighted mid over the full sample, clipped to ± 0.002 (± 20 bp). The bars at the two edges hold every return beyond the clip.

This quick pass did not yet exclude returns that straddle the maintenance break, a weekend, or a contract roll, so a few of its "one-minute" returns are really overnight or roll-to-roll gaps. That is why the kurtosis here (314) is higher than the 190 obtained for the same horizon in the cleaner analysis of the next section: a handful of gap returns of a fraction of a percent are enormous relative to a 3-basis-point typical move and inflate the fourth moment.

4. Return moments across horizons: mid versus microprice

4.1 Method

For each of nine sampling horizons – 30 s, 60 s, 2 min, 5 min, 15 min, 30 min, 1 h, 4 h and 1 day – the book state is loaded on that time grid, both the plain mid and the size-weighted mid are computed, and log returns of each are taken. Two filters are applied before computing any statistic:

- **Consecutive grid points only.** A return is kept only if the gap to the previous grid point is exactly one period. Any return that spans the maintenance break, a weekend or a holiday is dropped rather than treated as a one-period move.
- **No contract rolls.** A return whose two ends are on different contracts is dropped, so the price difference between expiring and new front months never enters.

The statistics are then annualised with the empirical number of periods per year (returns in sample ÷ years spanned), as explained in Section 1. Table 1 collects the results and Figure 4 shows the corresponding distributions.

Table 1 – Moments of ES log returns by sampling horizon, plain mid versus size-weighted mid (swmid). Consecutive grid points only, no contract rolls; annualised in trading time (empirical periods per year). Kurtosis is excess kurtosis; “zero returns” is the fraction of returns that are exactly zero.

horizon	price	n	periods/yr	mean (ann.)	variance (ann.)	vol (ann.)	skew	excess kurtosis	zero returns
30s	mid	2,594,165	533,550	+0.0061	0.02791	0.1671	+0.51	186.1	16.6%
30s	swmid	2,594,165	533,550	+0.0046	0.02804	0.1675	+0.72	198.5	0.1%
60s	mid	1,296,970	266,752	+0.0196	0.02786	0.1669	-0.06	186.5	11.6%
60s	swmid	1,296,970	266,752	+0.0086	0.02807	0.1675	+0.01	190.4	0.1%
2min	mid	649,405	133,566	+0.0155	0.02780	0.1667	-0.08	131.2	8.3%
2min	swmid	649,405	133,566	+0.0137	0.02801	0.1674	-0.00	132.7	0.0%
5min	mid	263,716	54,239	+0.0314	0.02782	0.1668	+0.20	120.7	5.2%
5min	swmid	263,716	54,239	+0.0228	0.02806	0.1675	+0.21	120.1	0.0%
15min	mid	90,139	18,539	+0.0199	0.02772	0.1665	+0.41	67.8	3.0%
15min	swmid	90,139	18,539	+0.0186	0.02791	0.1671	+0.43	67.2	0.0%
30min	mid	46,842	9,634	+0.0401	0.02734	0.1653	+0.02	32.0	2.3%
30min	swmid	46,842	9,634	+0.0341	0.02756	0.1660	+0.03	31.7	0.0%
1h	mid	24,183	4,974	+0.0557	0.02763	0.1662	+0.69	38.2	1.5%
1h	swmid	24,183	4,974	+0.0634	0.02800	0.1673	+0.68	37.2	0.0%
4h	mid	6,426	1,322	+0.0633	0.02721	0.1650	+0.80	33.4	0.7%
4h	swmid	6,426	1,322	+0.0632	0.02722	0.1650	+0.81	33.4	0.0%
1D	mid	1,015	209	+0.0961	0.02256	0.1502	+0.93	13.0	0.2%
1D	swmid	1,015	209	+0.0960	0.02256	0.1502	+0.93	13.0	0.0%

4.2 Volatility is flat across horizons; the mean is not measurable

The annualised volatility of the microprice is 16.75% at 30 seconds, 16.75% at one minute, 16.71% at 15 minutes, 16.73% at one hour and 16.50% at four hours – flat to within a fraction of a percentage point across a range of horizons spanning a factor of nearly 500. This is exactly what one expects if returns are (close to) uncorrelated in time: the variance of a return over k periods is k times the one-period variance, and dividing by the number of k -period intervals per year undoes the factor. A mild downward drift from 0.0281 at short horizons to 0.0272 at four hours would be consistent with slight negative autocorrelation of returns (bid-

ask bounce or short-term mean reversion), but the effect is at the 3% level in variance. The one-day figure of 15.0% is lower still, but with only 1,015 daily returns and an excess kurtosis of 13, the standard error of the daily variance estimate is around 12% of its value, so the daily figure is not significantly different from the rest. In round numbers, ES has run at about **16.7% annualised vol in trading time** over this five-year sample, which corresponds to a one-minute standard deviation of about 3.2 bp and a 30-second standard deviation of about 2.3 bp – close to one half-tick.

The annualised **mean** return, in contrast, rises steadily from about +0.5% at 30 seconds to +9.6% at one day. This is not a statement about drift at different horizons; it is an artefact of the consecutive-points filter interacting with where equity returns are earned. The short grids keep only intervals that lie wholly inside a continuous trading session, so they exclude every move across the maintenance break and every weekend – and a large part of the equity risk premium is realised in precisely those gaps (the well-documented "overnight return" effect). The one-day grid, sampling once every 24 hours, keeps the overnight sessions inside its returns and therefore captures most of the drift. Variance does not suffer the same distortion because the excluded gaps contribute relatively little variance compared with the drift they carry. In any case, the mean is not estimable here: the standard error of an annualised mean over 4.9 years is $\text{vol}/\sqrt{\text{years}} \approx 16.7\%/2.2 \approx 7.6$ percentage points, larger than every entry in the column. For a short-horizon volatility model the drift can be set to zero.

4.3 Kurtosis is huge at short horizons and still large at one day

Excess kurtosis is around 190 at 30 and 60 seconds, 130 at 2 minutes, 120 at 5 minutes, 67 at 15 minutes, 32–38 from 30 minutes to 4 hours, and 13 at one day. For comparison, a normal distribution has excess kurtosis 0; a Student-t with 5 degrees of freedom has 6. Two features stand out.

First, the decline with horizon is far slower than independence would predict. If one-period returns were independent and identically distributed with excess kurtosis 190, the excess kurtosis of a sum of n of them would be $190/n$; from 30 seconds to one trading day $n \approx 2,760$, so daily returns would be almost exactly normal (excess kurtosis 0.07). The observed 13 is two orders of magnitude larger. The gap is the signature of **volatility clustering**: the size of returns varies over time in a persistent way, so a daily return is a mixture of quiet-day and turbulent-day draws, and such a mixture has heavy tails even when each component is normal. The persistence that generates this is measured directly in Sections 8–10.

Second, the extreme kurtosis at 30–60 seconds, together with the fact that it barely changes between 30 s and 60 s, indicates that the tails at these horizons are driven by discrete jumps – single-second moves of many ticks (data releases, sweeps of the book) whose size does not scale with the horizon at all. Doubling the horizon roughly doubles the variance of the "diffusive" part but leaves a jump's contribution unchanged, so the tail-to-centre ratio hardly moves.

4.4 Mid versus size-weighted mid

The two price definitions give the same variance and the same kurtosis to within about 1% at every horizon; from four hours upward the two rows are essentially identical. They differ in three ways, all confined to short horizons:

- **Zero returns.** 16.6% of 30-second mid returns are exactly zero, falling to 11.6% at one minute, 5.2% at five minutes and 1.5% at one hour. The corresponding microprice fractions are 0.1% or below at every horizon. This is the half-tick quantisation described in Section

1: when the typical move is about the size of the grid step, the mid often does not move at all. A model fitted to mid returns at 30–60 seconds would be fitting a discrete distribution with a spike at zero; the microprice is the better series for a continuous model.

- **Slightly higher short-horizon variance for the microprice.** The swmid variance exceeds the mid's by about 0.5% (0.02804 versus 0.02791 at 30 s), because the microprice also moves when only queue sizes change. This is genuine information about the next price move, but it is also a small amount of extra "noise" that the plain mid does not have; the difference vanishes by four hours.
- **Skewness at the shortest horizons.** At 30 seconds the mid's skew is +0.51 and the microprice's +0.72; at 60 seconds they are -0.06 and $+0.01$. With excess kurtosis near 200, the sample skew is set by a handful of the largest moves and these numbers are not stable estimates of anything – the sign difference between 30 s and 60 s makes the point. From 15 minutes upward both series show a consistent positive skew of roughly +0.4 to +0.9, which is the more meaningful observation: in this sample the largest multi-hour and daily moves have been to the upside (the April 2025 relief rally being the obvious example), the reverse of the usual negative skew of equity index returns over longer samples.

4.5 The shape of the distribution

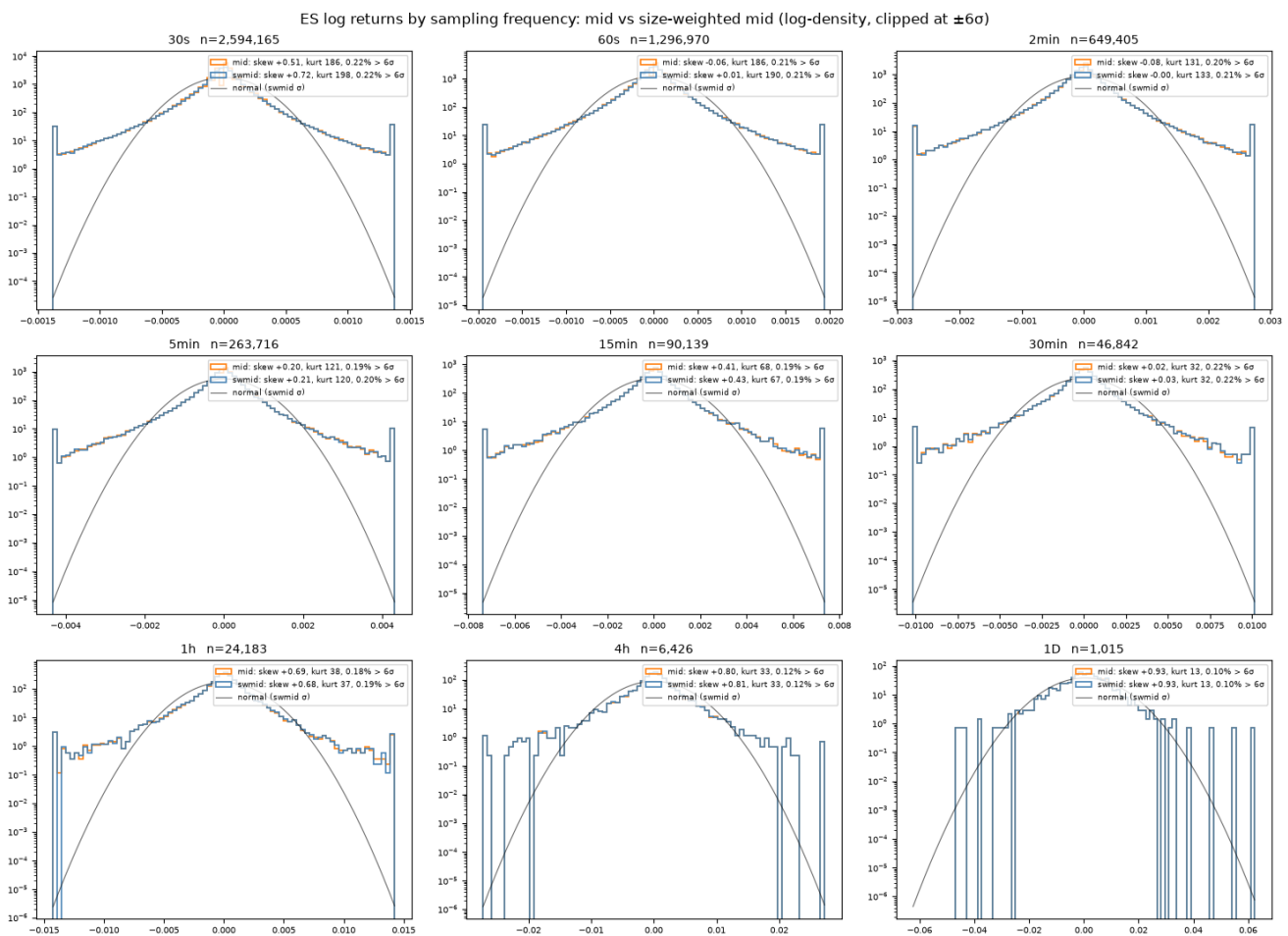


Figure 4. Log-density histograms of ES log returns at each horizon (orange: mid; blue: size-weighted mid), each overlaid with a normal density with the same mean and standard deviation as the microprice series (black). Each panel is clipped at ± 6 standard deviations; the mass beyond the clip is piled into the outermost bin, which is why every panel has a spike at each edge. Legends give skew, excess kurtosis and the fraction of returns beyond $\pm 6\sigma$.

Plotting density on a logarithmic axis is the standard way to look at tails, because a normal density becomes an inverted parabola and any excess above it is immediately visible. Figure 4 shows three things:

- The empirical histograms are **tent-shaped, not parabolic**, near the centre: on a log axis their sides are close to straight lines, which is the signature of a Laplace-like (exponential-tailed) distribution rather than a normal one. Beyond about two standard deviations they flatten out further into a broad shoulder that the normal curve is many orders of magnitude below – at 30 s to 5 min the empirical density at $\pm 5\sigma$ to $\pm 6\sigma$ is three to five orders of magnitude above the normal curve.
- The fraction of returns beyond $\pm 6\sigma$ is about **0.2% at every horizon from 30 seconds to 30 minutes** (roughly 1 in 500 returns), and still 0.1% at one day. A normal distribution puts 2×10^{-9} of its mass there. The edge spikes are therefore not a cosmetic artefact of the clipping: they represent a persistent population of moves that a normal model would call impossible.
- The mid and microprice histograms are **indistinguishable on this scale** at every horizon. The 16.6% zero-return mass of the 30-second mid falls inside a single central bin, so it does not show, and the tails of the two series are the same because a large move is a large move regardless of how the price is defined. At four hours and one day the histograms are too sparse (6,426 and 1,015 observations) to say much beyond "clearly heavier-tailed than normal".

5. Realised variance per minute from the one-second book

The sections that follow need a volatility measurement at one-minute resolution. A single sampled one-minute return is a poor measurement (its square has a standard error larger than itself), so instead a **per-minute realised variance** series is built from the one-second data: for each UTC minute, the sum of squared log returns of the microprice between consecutive snapshots that land in that minute, keeping only returns between rows at most two minutes apart and on the same contract. Alongside it are stored n , the number of one-second returns summed, and $moves$, the number of those in which the mid changed. Because the feed omits unchanged seconds, n averages 51.5 rather than 60. The series was built in one streaming pass over the 1.7 GB of Parquet (in nine seconds) and cached.

Table 2 – Summary of the per-minute series over 1,722,851 minutes (Nov 2021 – Sep 2026):
 rv = realised variance (sum of squared 1-second swmid log returns in the minute), n = number of 1-second returns in the minute, moves = number of those in which the mid changed.

series	count	mean	std	min	25%	50%	75%	max
rv	1,722,851	9.41e-08	2e-06	0	6.99e-09	2.09e-08	6.37e-08	0.000743
n	1,722,851	51.5	10.8	1	46	56	60	60
moves	1,722,851	18.3	13.1	0	8	16	27	60

The full series has 1,722,851 minutes with data, running from 00:00 UTC on 1 November 2021 to 21:00 UTC on 11 September 2026, summing 88.7 million one-second returns. The distribution of per-minute variance is very skewed: the median minute has $rv = 2.1 \times 10^{-8}$ (a per-minute standard deviation of 1.4 bp), the mean is 9.4×10^{-8} , four and a half times the median, and the maximum is 7.4×10^{-4} , thirty-five thousand times the median.

Summing all the per-minute variance and dividing by calendar years gives an annualised vol of **18.3%**, noticeably above the 16.7% of Table 1. Two effects account for the difference. About 0.43% of minutes fall inside the 17:00–18:00 ET break; they contain sparse pre-open indications with tiny sizes and wide or crossed books, and their mean variance is 29 times a normal minute's, so they contribute roughly 11% of the total variance despite being a negligible fraction of the time. Excluding them (as all subsequent sections do) brings the figure to 17.1%. The remainder is the well-known upward bias of realised variance at very fine sampling: one-second microprice returns contain some bid-ask bounce and queue noise that is negatively autocorrelated and partly cancels at 30-second sampling but not at one second. Neither effect is a concern for the intraday and persistence analysis, which is about relative variance across time.

6. Volatility by time of day and day of week

6.1 The intraday profile

All clocks in this section are US Eastern time so that US events line up across daylight-saving changes; the x-axis of Figure 5 runs in session order starting from the 18:00 ET Globex reopen. For each minute of the day the mean and the median per-minute variance across all days are computed and converted to annualised vol with the trading-time factor $252 \times 1,380$. Both are shown because they answer different questions: the mean is what matters for a

variance budget (it includes the rare huge minutes), while the median is what a typical minute at that time looks like. The gap between them is a measure of event risk at that time of day.

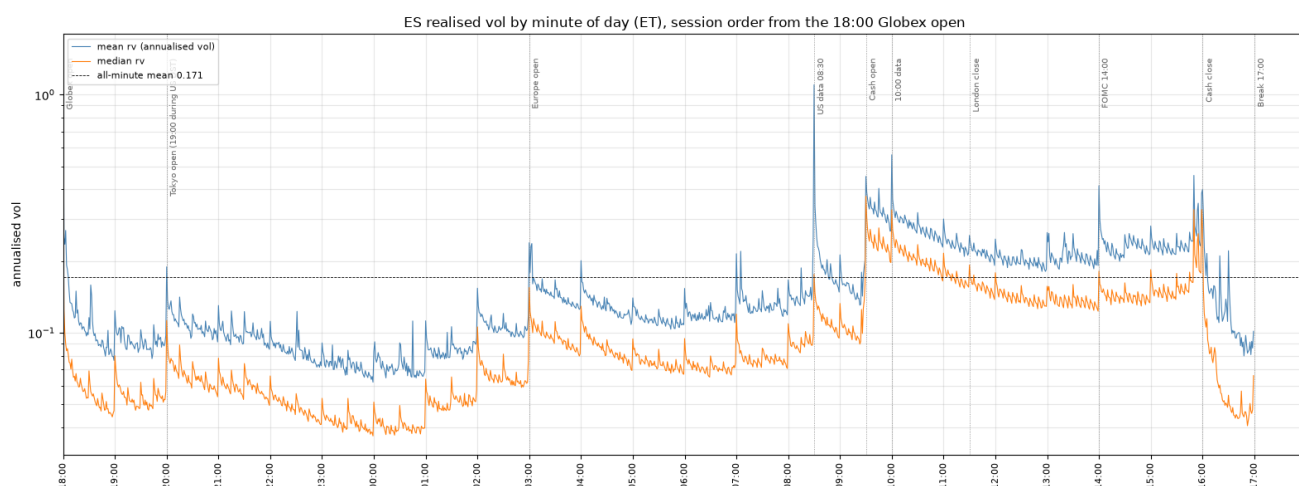


Figure 5. Annualised realised vol by minute of the day (ET), from the mean (blue) and the median (orange) per-minute variance over about 1,250 trading days, on a log scale. The dashed line is the all-minute mean (17.1%). Dotted vertical lines mark the session events labelled at the top.

The profile is a distorted "U" superimposed with sharp spikes. Reading it from left to right:

- The **18:00 Globex reopen** is the most violent minute of the entire day (annualised vol 149% from the mean, 59% from the median) as the book re-forms after the hour-long break, but it fades within minutes.
- The **Asian session** (roughly 19:00–03:00) is the quietest part of the day: mean-based vol drifts down from about 10% to 7% and median-based vol to about 4–5%, with a visible bump at the Tokyo cash open (20:00 ET during US summer time).
- The **European open** at 03:00 lifts vol by roughly a factor of two in a single step, to about 15–20%, followed by a slow decline through the European morning with smaller bumps at 02:00 (European pre-market) and 04:00 (continental cash open).
- The **08:30 US data release** minute is the second-largest spike (110% from the mean), and here the mean/median gap is at its widest – the median is only 18%. This is exactly the profile of a scheduled event: on most days nothing happens, on release days a great deal does.
- The **09:30 cash open** lifts the level of vol permanently for the day (to about 30–45%), with further spikes at **10:00** (a second data slot: ISM, consumer sentiment and similar) and at 09:31, 09:35 and 09:45. Vol then decays steadily through the US morning to a floor of about 20% around 13:00.
- The **14:00 FOMC slot** produces a spike (42%) on the eight days a year it is live, with a mean/median gap similar to 08:30's.
- Vol rises into the **16:00 cash close**, with the closing auction imbalance minutes (15:50, 15:59, 16:00) all among the top ten, then collapses through the post-close hour to about 9–10% before the 17:00 break.

The dashed line at 17.1% is exceeded only during the US session and at the opens: **most of the day, ES vol is well below its average**, and the average is made up by a few concentrated hours.

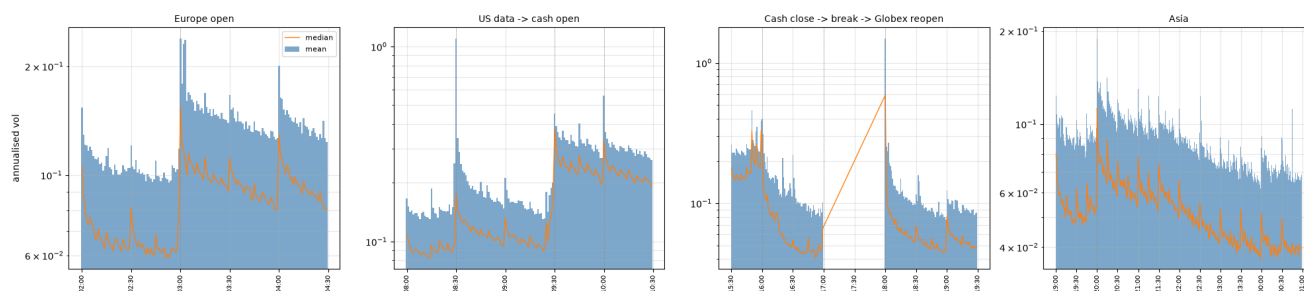


Figure 6. Minute-resolution zooms on four parts of the day (mean as bars, median as line, log scale): the European open, the run from the 08:30 data release through the cash open, the close/break/reopen sequence (the straight orange segment across 17:00–18:00 is just the plot connecting two points across the excluded break), and the Asian session.

The zooms make the structure of the opens clear. The European open is a step change with a slow fade. The 08:30 spike is a single minute, then vol drops back to almost the pre-release level within about ten minutes before the cash open lifts everything again. The cash open, by contrast, is not a spike but a level shift: vol steps up at 09:30 and stays up. The close/reopen panel shows the post-close decay, the empty break, the reopen spike and the second, gentler fade into the Asian night; the Asian panel shows the near-monotone decline through the night interrupted by the Tokyo open.

Table 3 – The twelve highest-variance minutes of the day (US Eastern clock), ranked by mean realised variance and expressed as annualised vol.

minute (ET)	ann. vol from mean rv	ann. vol from median rv	days observed
18:00	1.493	0.591	1,256
08:30	1.100	0.176	1,258
10:00	0.559	0.328	1,236
15:50	0.458	0.329	1,193
09:30	0.453	0.374	1,256
14:00	0.415	0.182	1,194
09:45	0.404	0.276	1,236
16:00	0.399	0.277	1,203
09:31	0.390	0.305	1,241
15:59	0.386	0.329	1,202
09:35	0.371	0.272	1,236
10:01	0.363	0.271	1,236

Table 3 lists the twelve minutes with the highest mean variance. Nine of the twelve are opens, the minutes just after an open, or scheduled release slots; the other three are the closing auction. The 18:00 reopen has the largest *median* as well as mean, so its violence is routine rather than event-driven; 08:30's median is six times smaller than its mean and 14:00's more than twice, confirming their event-driven nature.

6.2 Where the variance goes: session budget

Table 4 – Where the day's variance happens. “Share of variance” is the fraction of total realised variance falling in the window; “variance density” is that share divided by the window's share of minutes, so 1.00 is an average minute.

session	minutes/ day	days	ann. vol (mean rv)	ann. vol (median rv)	share of variance	share of minutes	mid changes / min	variance density
Globex reopen 18:00-19:00	60	1,256	0.228	0.060	7.8%	4.4%	12.7	1.78
Asia 19:00-03:00	480	1,515	0.092	0.053	10.1%	35.2%	12.3	0.29
Europe 03:00-08:30	330	1,258	0.132	0.082	14.5%	24.2%	17.8	0.60
US pre-open 08:30-09:30	60	1,258	0.227	0.109	7.8%	4.4%	21.4	1.77
Cash open 09:30-10:30	60	1,256	0.320	0.233	15.2%	4.3%	36.3	3.51
US midday 10:30-14:00	210	1,236	0.221	0.152	25.1%	15.0%	26.9	1.67
US afternoon 14:00-16:00	120	1,221	0.241	0.152	16.6%	8.3%	26.4	1.99
Post-close 16:00-17:00	60	1,218	0.143	0.061	2.9%	4.2%	11.8	0.70

Table 4 turns the profile into a budget. The ranking by **variance density** (share of variance divided by share of minutes) is unambiguous: the cash-open hour 09:30–10:30 runs at 3.5 times an average minute and alone carries 15.2% of the day's variance; the US afternoon 14:00–16:00 runs at 2.0×; the Globex reopen hour, the 08:30–09:30 pre-open hour and the US midday all run at about 1.7–1.8×. The whole of the US regular session (09:30–16:00, 6.5 hours) accounts for about 57% of daily variance in 28% of the minutes. At the other end, the eight-hour Asian session carries just 10% of the variance at a density of 0.29 – a typical Asian minute has less than a third of the average minute's variance and less than a tenth of a cash-open minute's. The "mid changes per minute" column tells the same story in terms of activity: the mid ticks about 36 times a minute in the cash-open hour and 12 times a minute overnight.

A practical consequence for any model that predicts the probability of a move over the next few minutes: **the time of day is worth a factor of ten or more in variance**, before anything else is considered. A model that ignores it is wrong by that factor most of the day.

6.3 Day of week

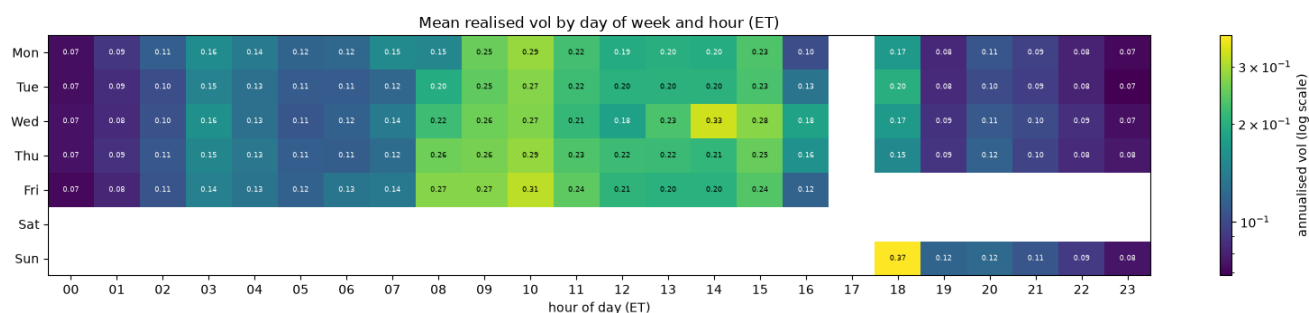


Figure 7. Mean realised vol (annualised, log colour scale) by day of week and hour of day (ET). The 17:00 column is empty (the break); Saturday is empty; Sunday has only the 18:00–24:00 evening session; Friday has no evening.

Table 5 – Annualised vol by weekday (Monday–Friday), from the mean and the median per-minute variance.

weekday	ann. vol from mean rv	ann. vol from median rv
Mon	0.1596	0.0786
Tue	0.1606	0.0840
Wed	0.1766	0.0848
Thu	0.1708	0.0888
Fri	0.1887	0.1050

The heatmap shows that the intraday pattern is essentially the same every weekday, and that any day-of-week effect is modest next to the time-of-day effect. The Sunday 18:00 reopen hour (37%) is the hottest cell in the grid, larger than the weekday reopens (15–20%), because it absorbs a whole weekend of news. Wednesday 14:00 (33%) stands out because FOMC decisions land on Wednesdays. Friday is the most volatile weekday by both mean (18.9%) and median (10.5%), with the 08:30 slot a shade hotter on Fridays (27%) than other days – the monthly payrolls release. Monday and Tuesday are the quietest (about 16.0% mean). The spread between the quietest and busiest weekday is about 18% in vol terms, a second-order effect that a model could include as a small multiplier.

7. How fast does an open's extra volatility fade?

For each of four "opens" (Globex 18:00, Europe 03:00, the 08:30 data release and the 09:30 cash open) and the 16:00 cash close, Figure 8 plots the mean per-minute variance at k minutes after the event, as annualised vol, for k from 0 to 180. A curve of the form $v(k) = b + a \cdot \exp(-k/\tau)$ is fitted on minutes 1–120: b is the plateau the vol settles to, a the initial excess above it and τ the e-folding time, so that the half-life of the excess is $\tau \cdot \ln 2$ and the time for it to fall below 10% of the plateau is $\tau \cdot \ln(10a/b)$.



Figure 8. Mean realised vol (annualised) in the 180 minutes after each event, log scale. Dashed lines are the plateau-plus-exponential fits on minutes 1–120. The fits for 08:30, 09:30 and 16:00 are not meaningful, for reasons given in the text.

Table 6 – Exponential-plus-plateau fits $v(k) = b + a \cdot \exp(-k/\tau)$ to the mean per-minute variance in minutes 1–120 after each event. Vol columns are annualised. Only the Globex and Europe rows are well-posed fits – see the discussion.

event	vol at $k=0$	vol at $k=1$	plateau vol b	excess var at $k=1$ / plateau	tau (min)	half-life (min)	mins to <10% excess
Globex open 18:00	1.493	0.252	0.096	6.399	4.952	3.432	21.593
Europe open 03:00	0.239	0.179	0.133	1.326	10.146	7.033	27.225
US data 08:30	1.100	0.340	0.502	-0.908	263.509	182.650	582.328
Cash open 09:30	0.453	0.390	0.005	4584.949	125.705	87.132	1350.210
Cash close 16:00	0.399	0.312	2.085	-1.038	600.000	415.888	1405.085

Two of the five fits are clean and informative. After the **Globex reopen**, variance drops from an annualised 149% at $k = 0$ to 25% at $k = 1$ and then decays with $\tau \approx 5$ minutes (half-life 3.4 minutes) to a plateau of 9.6%; the excess is below 10% of the plateau after about 22 minutes. After the **European open**, the initial excess is smaller (24% at $k = 0$ against a plateau of

13.3%) and decays with $\tau \approx 10$ minutes (half-life 7 minutes), reaching the 10% threshold after about 27 minutes. In both cases the extra variance of the open is essentially gone within half an hour, and the first few minutes carry most of it.

The other three fits should be disregarded, and the reasons are instructive. The **08:30** window contains the 09:30 cash open at $k = 60$, which is a larger and more persistent lift than the release spike being fitted, so the fit returns a negative excess and a rising curve. The **09:30** fit produces a plateau of essentially zero (0.5%) with a τ of two hours: it is not describing a decay to a level but the slow, roughly exponential slide of vol through the US morning, which has no plateau within the window – the cash open is a regime, not a shock. The **16:00** fit hits the upper bound placed on τ (600 minutes) and its "plateau" of 209% is a numerical artefact: the fitting window runs across the maintenance break into the 18:00 reopen, which sits at $k = 120$, and the curve is trying to bend upward towards it. The qualitative reading from the plot is nevertheless clear: after the close, vol steps down and continues to fall for the whole post-close hour with no plateau.

8. Autocorrelation of volatility

8.1 Method: deseasonalise, then work in trading time

The intraday profile of Section 6 is deterministic – it repeats every day – so any autocorrelation computed on raw per-minute variance would mostly rediscover it (there would be a peak at every multiple of one day). To measure the persistence of volatility *beyond* the daily cycle, the series is first **deseasonalised**: $x = rv / s(\text{tod})$, where $s(\text{tod})$ is the mean variance at that minute of the day. By construction x has mean 1, and 1 is the "baseline" – $x = 3$ means the minute had three times the variance normal for that time of day. Minutes with fewer than 20 one-second returns or zero variance are dropped, leaving 1,698,896 usable minutes (99.0%); their mean x is 1.008 and the standard deviation of $\log x$ is 1.28.

The autocorrelation is computed on **log x** rather than x itself. Log realised variance is close to normally distributed, whereas the ratio x has extremely heavy tails (its maximum is in the thousands) that make sample autocorrelations noisy and biased downward; the orange line in Figure 9 shows how much lower and rougher the ACF of the raw ratio is. Lags are counted in **trading minutes**: the series is the sequence of consecutive tradeable minutes with the break and the weekends removed, so a lag of 1,380 is "the same time yesterday" and 4,140 is three trading days.

Three functional forms are fitted to the ACF, using log-spaced lags so that the thousands of long lags do not swamp the few short ones:

- **Exponential**, $a \cdot \exp(-k/\tau)$: a single decay rate, which is what a GARCH(1,1) or any first-order autoregressive model of volatility implies. On a semi-log plot it is a straight line.
- **Power law**, $c \cdot k^{-\alpha}$: "long memory" – the autocorrelation halves every time the lag is multiplied by a fixed factor, so there is no characteristic time scale. On a log-log plot it is a straight line. This is what studies of daily realised variance usually find.
- **Two exponentials**, $a_1 \cdot \exp(-k/\tau_1) + a_2 \cdot \exp(-k/\tau_2)$: a fast component (news, microstructure) plus a slow component (regime), which is the picture behind the HAR model of Section 10.

8.2 One-minute realised variance, three days of lags

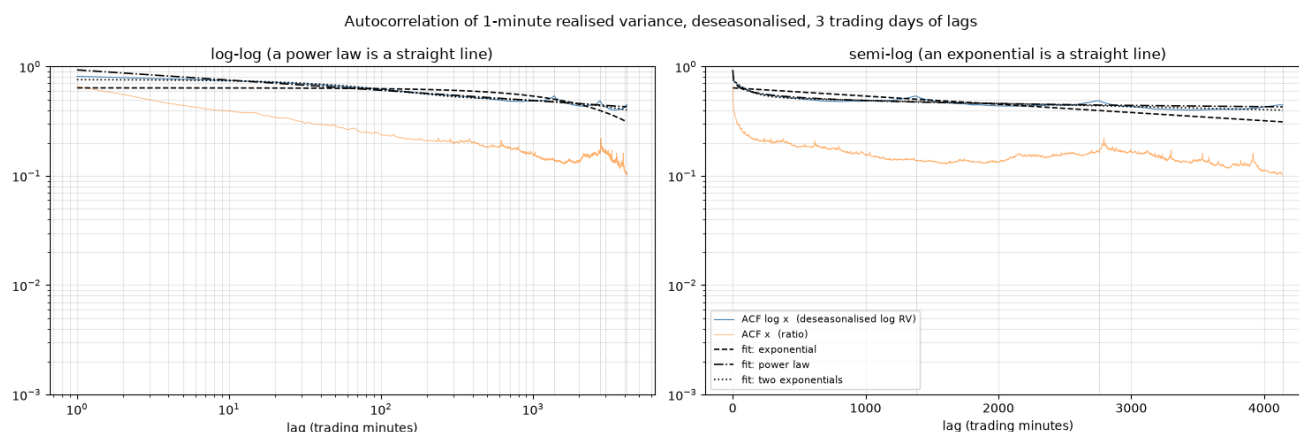


Figure 9. Autocorrelation of deseasonalised one-minute log realised variance (blue) and of the raw ratio (orange) out to 4,140 trading minutes, on log-log axes (left, where a power law is straight) and semi-log axes (right, where an exponential is straight). Black lines are the three fits; dotted verticals mark one, two and three trading days.

Table 7 – Fits to the autocorrelation function of deseasonalised 1-minute log realised variance, lags 1 to 4,140 trading minutes (three trading days). Parameters are listed in the order a , τ (exponential); c , α (power law); a_1 , T_1 , a_2 , T_2 (two exponentials), τ in minutes.

model	R^2 (log-spaced lags)	fitted parameters		implied half-life
exponential	0.637	0.6372, 5783		4008.4 min
power law	0.961	0.9302, 0.09339	$\alpha=0.093$: rho halves every x1672.9 in lag	
two exponentials	0.984	0.2373, 114.6, 0.5222, 1.541e+04	fast 79.5 min (weight 31%), slow 10681 min	

The autocorrelation of $\log x$ is 0.81 at one minute, 0.77 at five minutes, 0.70 at 30 minutes, 0.65 at one hour, 0.52 at 390 minutes (a full cash session) and 0.54 at one trading day. That last pair is the key observation: **after the first hour the ACF barely falls at all**. Half of the deviation of a minute's variance from its time-of-day norm is still present a day later, and the slight rise from 390 to 1,380 minutes (visible as small bumps at multiples of a day in the plot) is residual daily seasonality that the mean-by-minute normalisation did not entirely remove – days that are volatile at 10:00 tend to be volatile at 10:00 the next day as well.

The single exponential is a poor description ($R^2 = 0.64$): its fitted half-life of 4,000 minutes is a compromise between the fast initial drop and the near-flat tail, and it fits neither. The power law does well ($R^2 = 0.96$) with a very small exponent, $\alpha = 0.093$, meaning the ACF halves only every $\sim 1,700$ -fold increase in lag – so slow that it is almost a constant. The two-exponential fit does best ($R^2 = 0.98$) and gives an interpretable decomposition: a **fast component with a half-life of 80 minutes carrying 31% of the weight**, and a **slow component with a half-life of about 10,700 minutes – nearly eight trading days – carrying 69%**. The two amplitudes sum to 0.76, so the remaining 24% of the variance of $\log x$ is unpredictable minute-to-minute noise, consistent with the estimation error of a variance measured from about 50 one-second returns.

8.3 Hourly realised variance, thirty days of lags

To see whether the slow component is real at the horizon that matters for a day-ahead view, the analysis is repeated on hourly realised variance (sum of the per-minute rv over each hour, requiring at least 600 one-second returns, i.e. at least ten minutes of data), deseasonalised by hour of day in logs, with lags out to 690 trading hours (30 trading days at 23 hours per day).

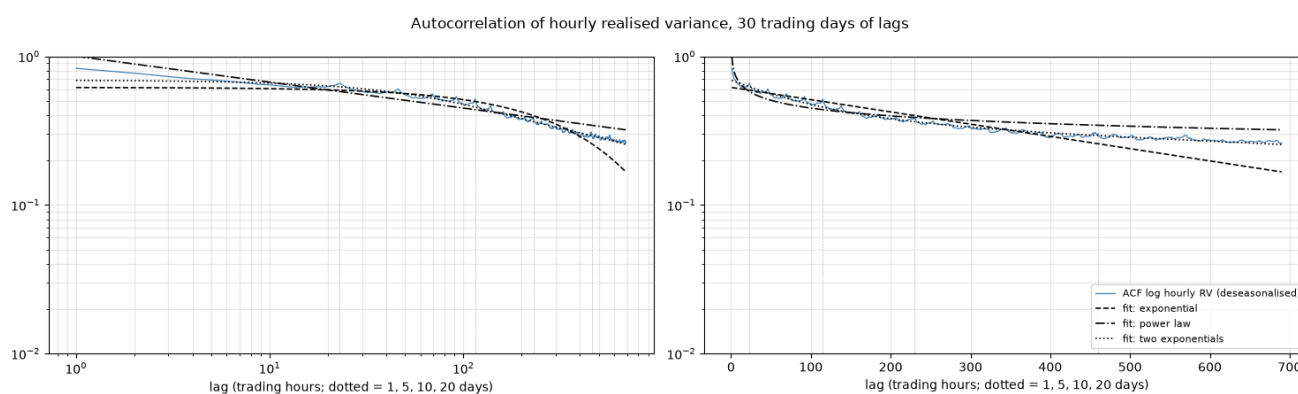


Figure 10. Autocorrelation of deseasonalised hourly log realised variance to 690 trading hours (30 trading days), with the same three fits; dotted verticals mark 1, 5, 10 and 20 trading days.

Table 8 – The same three fits to the autocorrelation of deseasonalised hourly log realised variance, lags 1 to 690 trading hours (30 trading days); τ in hours.

model	R^2 (log-spaced lags)	fitted parameters		implied half-life
exponential	0.890	0.6173, 526.7		365.1 h
power law	0.878	1.001, 0.1746	$\alpha=0.175$: rho halves every x53.0 in lag	
two exponentials	0.977	0.3183, 103.6, 0.373, 1803	fast 71.8 h (weight 46%), slow 1250 h	

The hourly ACF is 0.83 at one hour, 0.66 at one day, 0.52 at five days, 0.39 at ten days and 0.30 at twenty days. Here the picture is different from the minute scale in an important way: on the log-log panel the ACF is *concave* – it falls away faster and faster at long lags – rather than straight, so the pure power law now fits slightly worse than the exponential (R^2 0.88 versus 0.89) and clearly worse than the two-exponential (R^2 0.98). The long memory seen over three days is, at this scale, resolved into a **fast component with a half-life of 72 hours (about three trading days) carrying 46% of the weight** and a **slow component with a half-life of 1,250 hours (about 54 trading days, i.e. two and a half months) carrying 54%**. The single-exponential half-life of 365 hours (16 trading days) is again a poor compromise between them.

The two ACFs are consistent with each other and with the standard picture of equity index volatility: a cascade of time scales – minutes-to-hours (the 80-minute component), days (the 3-day component) and months (the 54-day component) – rather than either one half-life or a true scale-free power law. For practical purposes, **any forecast of ES vol at horizons of minutes to a day should condition on at least three windows of history: the last hour or so, the last few days, and the last couple of months**. This is precisely the HAR structure of Section 10.

9. Response to a volatility shock

The autocorrelation is an average over all minutes. A more direct question for a trading model is: given that the market has just had an unusually violent minute, what is the expected variance over the following minutes and hours? Shock minutes are defined as those whose deseasonalised variance x exceeds the 99th percentile ($x > 9.1$) or the 99.9th percentile ($x > 36.5$). They are split into **scheduled** shocks – those at 08:30, 10:00 or 14:00 ET or the minute after – and **unscheduled** ones, everything else. For each set, the average of x at every offset k from 60 minutes before to 720 minutes after the shock is computed; since x has mean 1, the quantity $E[x_{t+k} | \text{shock}] - 1$ is the excess variance ratio, plotted on a log scale with exponential and power-law fits.

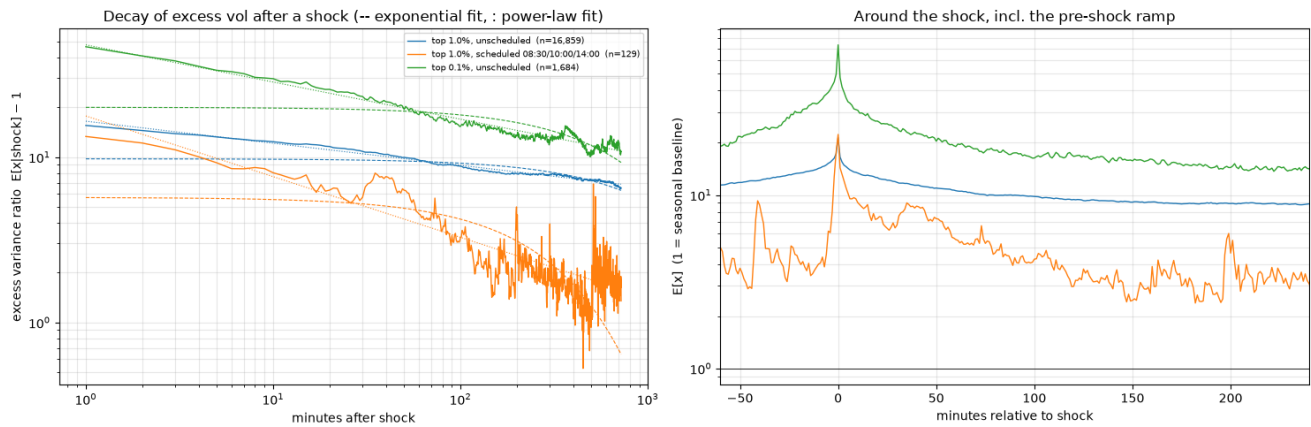


Figure 11. Left: excess variance ratio $E[x | \text{shock}] - 1$ against minutes after the shock, log-log, with exponential (dashed) and power-law (dotted) fits. Right: $E[x]$ from 60 minutes before to 240 minutes after the shock, log scale; the horizontal line at 1 is the seasonal baseline. Blue: top 1% unscheduled; orange: top 1% scheduled; green: top 0.1% unscheduled.

Table 9 – Mean deseasonalised variance ratio x around the shock minute ($k = 0$). $x = 1$ is the time-of-day baseline. The notebook's "minutes until excess < 10%" column is omitted: it was undefined ("—") for every group because the excess never fell that far within the 720-minute window.

	shock set	n	threshold (x)	x at k=0	x at k=-1	x at k=1	x at k=10	x at k=60	x at k=390	exp. fit half-life (min)	power- law α
	top 1.0%, unscheduled	16,900	9.11	21.4	17.3	16.5	13.4	10.6	8.59	1130	0.133
	top 1.0%, scheduled 08:30/10:00/14:00	129	9.11	22.3	17.8	14.3	9.04	5.53	3.04	228	0.367
	top 0.1%, unscheduled	1,680	36.5	73.4	49.7	47.5	30.8	19	15.1	650	0.225

The most striking feature of the right-hand panel is the **pre-shock ramp**. In the minute *before* a top-1% unscheduled shock the variance ratio is already 17.3 (versus 21.4 at the shock itself), and an hour before it is still about 12. For the top-0.1% set the corresponding numbers are 49.7 before and 73.4 at the shock. In other words, an "unscheduled shock" in this sample is almost never a bolt from the blue: it is the largest minute inside a stretch that was already running at ten to twenty times its normal variance. Extreme minutes cluster in turbulent regimes. This is the shock-conditional version of the long memory in Section 8.

The consequence is that the decay after an unscheduled shock is **extremely slow**. For the top-1% set the variance ratio is 16.5 one minute after, 13.4 ten minutes after, 10.6 an hour after and **still 8.6 after 390 minutes** (a full cash session). At no point in the 720-minute window does the excess fall below 10% of baseline – the notebook's "minutes to <10% excess" column is undefined for every group. The exponential fit has a half-life of 1,130 minutes (nearly a trading day) and is visibly wrong at short lags, where it is flat; the power law with $\alpha = 0.13$ tracks the curve much better, and an exponent that small again means "no characteristic time scale within the window". For the top-0.1% set the decay is steeper ($\alpha = 0.23$, exponential half-life 650 minutes) but from a much higher level, and after 390 minutes the variance ratio is still 15.

The 129 **scheduled** shocks behave differently and as one would hope: the variance ratio is 14.3 one minute after, 9.0 after ten minutes, 5.5 after an hour and 3.0 after 390 minutes, with a power-law exponent of 0.37 (nearly three times the unscheduled one) and an exponential half-life of 228 minutes. The pre-shock level is also much lower (about 3–5 in the hour before the release; the bump around –30 to –40 minutes is the 09:30 cash open preceding the 10:00 releases), confirming that the market was not already turbulent. Note that even a scheduled release leaves vol at three times normal six hours later – the number is only "fast" by comparison with the unscheduled case. Note also that scheduled shocks, though only 129 of

the roughly 17,000 top-1% minutes (the sample of top-0.1% scheduled minutes was too small to analyse), are highly predictable in time and should be handled by a calendar rather than by a persistence model.

10. A HAR model on hourly realised variance

The Heterogeneous AutoRegressive (HAR) model of Corsi (2009) is the workhorse of realised-variance forecasting. It regresses future log variance on three simple averages of past log variance – over the last period, the last "day" and the last "week" – which is a parsimonious way to capture the cascade of time scales found in Section 8. Here it is fitted to deseasonalised hourly log realised variance in trading time, with the day defined as 23 trading hours and the week as 115:

$$y_{[t+1..t+h]} = \beta_0 + \beta_h \cdot y_t + \beta_d \cdot \text{mean}(y_{t-22..t}) + \beta_w \cdot \text{mean}(y_{t-114..t}) + \varepsilon$$

where the target is the mean log variance over the next h hours, for $h = 1, 4, 23$ (a day) and 115 (a week). Each is compared with a plain AR(1) (last hour only) and, implicitly, with the seasonal forecast $y = 0$ that an R^2 of zero would correspond to. Coefficients are fitted on the first 70% of the sample and the out-of-sample R^2 is computed on the final 30%.

Table 10 – HAR and AR(1) regressions of future mean log realised variance (deseasonalised, hourly, trading time) on the last hour, the last-day mean and the last-week mean. Out-of-sample R^2 is on the final 30% of the sample with coefficients fitted on the first 70%.

horizon	model	R^2 in-sample	R^2 out-of-sample	β last hour	β last day (23 h mean)	β last week (115 h mean)
1 hour	HAR	0.746	0.693	0.543	0.352	0.084
1 hour	AR(1)	0.703	0.650	0.839	–	–
4 hours	HAR	0.793	0.732	0.369	0.464	0.135
4 hours	AR(1)	0.699	0.637	0.773	–	–
1 day	HAR	0.826	0.736	0.181	0.502	0.253
1 day	AR(1)	0.649	0.564	0.681	–	–
1 week	HAR	0.755	0.573	0.106	0.445	0.296
1 week	AR(1)	0.560	0.407	0.592	–	–

Three conclusions follow from Table 10.

- **Vol is substantially forecastable from its own history alone.** The HAR explains 69% of the variance of next-hour log RV out of sample, 73% of next-four-hour, 74% of next-day and 57% of next-week. These are large numbers – they are the ceiling for what a persistence model can do without any external information such as the economic calendar or implied vol. The out-of-sample R^2 is only 5–9 points below the in-sample one at horizons up to a day, so the fit is not over-parameterised; at the one-week horizon the gap widens to 18 points, as expected with fewer effectively independent weeks.
- **The HAR beats the AR(1) at every horizon, and by more at longer horizons:** 4 points of out-of-sample R^2 at one hour, 10 at four hours, 17 at one day and 17 at one week. The AR(1)'s single coefficient (0.84 at one hour) cannot simultaneously represent the fast decay of the last hour's information and the slow persistence of the regime.
- **The weight shifts from the last hour to the last day and week as the horizon lengthens.** For the one-hour target the coefficients are 0.54 on the last hour, 0.35 on the last day and 0.08 on the last week; for the one-day target they are 0.18, 0.50 and 0.25; for one week 0.11, 0.45 and 0.30. The last hour is the best single predictor of the next hour, but for anything beyond a few hours it is the last day's average that matters most, and the

two-month regime component appears as the weekly term. The coefficients at the one-hour horizon sum to 0.98, close to one, which is the HAR's way of expressing the near-unit-root persistence seen in the ACF.

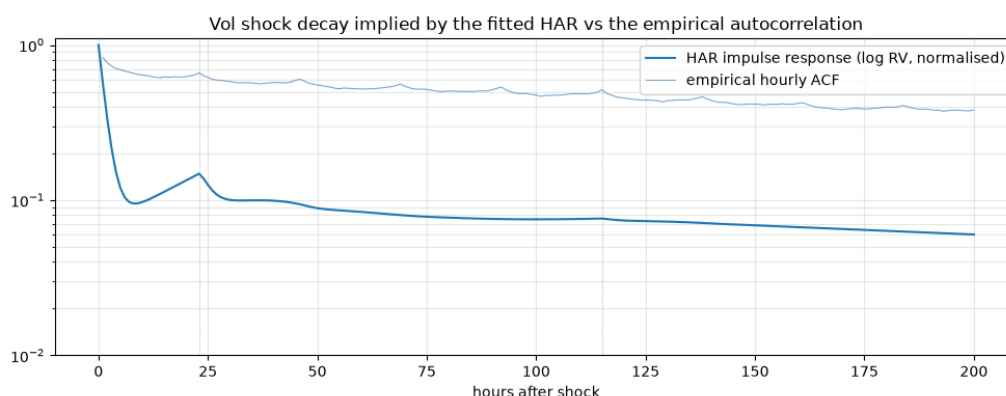


Figure 12. Response of the fitted one-hour HAR to a one-standard-deviation shock to the "last hour" term, iterated forward (thick line, normalised to 1 at hour 0), against the empirical hourly ACF (thin line). Log scale; dotted verticals at 1 and 5 trading days.

Figure 12 iterates the fitted one-hour HAR forward after a single one-hour shock. The implied response halves in 2 hours and falls to 10% after 7 hours, with a small secondary bump at 23 hours when the shocked hour drops out of the daily average. This is far faster than the empirical ACF, which is still at 0.66 after a day, and the notebook's own note that the two "should track each other if the model is adequate" needs qualifying. The two curves answer different questions. The impulse response is the effect of a shock confined to *one hour* with the previous day and week at their baseline – and the HAR quite correctly says such an isolated hour is mostly forgotten by the next morning, because its $1/23$ contribution to the daily mean and $1/115$ to the weekly mean is small. The ACF, on the other hand, is the average co-movement of two hours a given distance apart, and Section 9 showed that a high-variance hour almost always sits inside a high-variance day and week; it inherits the persistence of those slower components. The impulse response and the ACF would coincide only for an AR(1), where a single lag carries all the history. So the comparison does not indicate that the HAR is inadequate; the R^2 values in Table 10 are the relevant test, and they are good.

11. What this means for modelling

The purpose of this study is to inform a model of short-horizon (minutes) price variability, ultimately for pricing five-minute up/down markets. The findings translate into a fairly specific set of modelling requirements.

1. **Use the microprice, not the mid, at horizons below a few minutes.** Both give the same variance, but the mid is a discrete variable with a spike at zero (17% of 30-second returns) and cannot be modelled with a continuous distribution. Above 15 minutes the choice does not matter.
2. **Drift is zero.** The annualised mean return cannot be distinguished from zero at any horizon (standard error of about 7.6 percentage points), and the monotone rise of the estimate with horizon is a filtering artefact. For a five-minute window the expected move is 0 for all practical purposes; the entire problem is the variance.
3. **Time of day is the first-order variance model, and it is worth a factor of ten.** A deterministic minute-of-day profile $s(\text{tod})$, estimated as in Section 6, captures the difference between an Asian-session minute (variance density 0.3) and a cash-open minute (3.5), the exact minutes of the opens and the scheduled release slots. It should be the base

of any model; a weekday multiplier (Friday about 18% higher vol than Monday, Sunday reopen and Wednesday 14:00 hotter) is a small refinement.

4. **Opens are short-lived, scheduled releases are single minutes, but the cash open is a level shift.** The excess variance of the Globex and European opens has a half-life of 3–7 minutes and is gone within half an hour; the 08:30 and 14:00 spikes last a minute or two on the days they are live. A model needs the calendar of release days to know when the 08:30 and 14:00 minutes will be violent, since the median for those minutes is a small fraction of the mean. The 09:30 open is different: it does not decay to a plateau but starts a slow slide through the morning.
5. **On top of the seasonal profile, condition on recent realised variance at three time scales.** Deseasonalised vol has an 80-minute fast component, a ~3-day component and a ~2-month regime component, and a HAR on hourly log RV with last-hour/last-day/last-week terms recovers about 70% of the out-of-sample variance of the next hour's log variance. A single-half-life GARCH-type model is a poor fit at any horizon studied. The natural structure is multiplicative: $\text{variance} = s(\text{tod}) \times \text{weekday factor} \times \exp(\text{HAR forecast of log } x)$.
6. **Do not expect vol to revert quickly after a large move.** Six and a half hours after a top-1% unscheduled minute, expected variance is still 8–9 times its time-of-day norm; after a top-0.1% minute it is 15 times. Equivalently, a violent minute is much more likely to be followed by more violent minutes than by calm – the pre-shock ramp shows such minutes almost always arrive in already-turbulent stretches. A model should ramp its variance up sharply on a large realised move and bring it down only slowly, more slowly than any single exponential.
7. **Tails are heavy at every horizon and a normal is inadequate for anything but the centre.** About one return in 500 is beyond six standard deviations at horizons from 30 seconds to 30 minutes, versus one in 500 million for a normal, and the excess kurtosis is around 190 at one minute. Some of this is absorbed by the conditional-variance model (a "6 σ " move at 08:30 on a payrolls day is not 6 σ conditionally), but the residual conditional distribution will still need heavy tails – a Student-t or a mixture with a jump component – to price the probability of a large five-minute move sensibly.
8. **Quote the unconditional numbers in trading time.** The 16.7% annualised vol is per trading time and corresponds to about 3.2 bp per minute and 7 bp per five minutes on average; but "on average" is close to meaningless intraday, since the same five minutes at 02:00 ET and at 09:31 ET differ by a factor of ten in variance.

12. Caveats and suggested next steps

- **Microstructure noise in one-second realised variance.** The per-minute rv series is built from one-second microprice returns and is biased upward by bid-ask bounce and queue noise (18.3% versus 16.7% annualised). This does not affect the shape of the intraday profile or the persistence measurements, but the absolute level should be calibrated against a coarser (30-second or one-minute) estimator, or a noise-robust one, before it is used as a variance forecast.
- **Residual seasonality after deseasonalisation.** The ACF of one-minute log RV shows small bumps at multiples of one trading day, so the mean-by-minute normalisation leaves some day-to-day structure in place. Normalising in logs, or by a smoothed profile, and checking for a weekday interaction would clean this up.

- **The decay-after-open fits for 08:30, 09:30 and 16:00 are degenerate** because the 120-minute fitting window contains a second event. Shorter windows (for 08:30, minutes 1–50) or fitting the decay of the deseasonalised ratio rather than the raw variance would give usable numbers.
- **Only 129 scheduled shocks** reached the top-1% threshold, so the scheduled-shock curve is noisy after the first hour. Conditioning on the release calendar directly (all release days, not only those that produced a top-1% minute) would give a far larger and unbiased sample.
- **Skewness estimates at horizons below 15 minutes are not meaningful** with excess kurtosis near 200; the positive skew at 15 minutes to one day is a property of this particular five-year sample and should not be built into a model as a structural feature.
- **Natural extensions:** fit the multiplicative model of point 5 above at one-minute resolution and evaluate its out-of-sample five-minute variance forecasts directly; add implied volatility (VIX or ES options) and the economic calendar as regressors; and repeat the whole analysis on the BTC/USD series that the target markets settle on, to establish whether the ES structure – flat vol across horizons, a ten-fold intraday range, an 80-minute/3-day/2-month persistence cascade – carries over to a 24/7 asset with no cash open.